

Network discovery questionnaire

Network team discovery questionnaire

Engagement: **Stabilize network** for existing **RHEL on Azure** workloads — document as-built, identify blockers, recommend controls. **Not** build hub/spoke or network IaC in client tenant.
Master scope: ENGAGEMENT-ALIGNMENT.md.

Purpose: Track B discovery — network requirements and gaps for **RHEL + Podman** (Snowflake / data platform), aligned with SOW.md (contract items **1–6**), CHARTER-ADDENDUM.md, and STRATEGY-POST-RHEL.md.

Audience: Client network / cloud networking / security architecture teams.

How to use: Use only questions marked **In scope**, **Client-owned**, or **Advisory**. **Out of scope** = consultant does not implement.

Scope legend

Tag	Meaning
In scope	Consultant documents answers and drives recommendations, standards, risks, cutover guidance, or KT per SOW.
Client-owned	Must be asked in discovery; client executes or decides . Consultant does not implement or own outcome.
Advisory	Consultant may recommend or draft standards/runbooks; client implements in production.
Out of scope	Do not expand engagement to cover; redirect to client app, GRC, ops, or separate SOW.

SOW quick reference (contract items 1–6)

SOW item	Contract scope	Track B in this engagement
1	Migration planning and technical guidance	As-built review, RACI, environment scope
2	RHEL standards & best practices	Network controls for RHEL workloads (draft)
3	Risk identification and mitigation	Network risk register entries
4	Environment improvement recommendations	PE, build spoke, egress recommendations
5	Cutover support	Network runbook draft ; support in agreed window
6	Knowledge transfer & support planning	KT on network runbooks; post-KT ownership

In scope (consultant)	Out of scope (client / change order)
Network requirements , allow-list draft , risks, runbook drafts	Firewall/peering/PE implementation , hub greenfield, network Terraform in client sub, SOC deploy, audit sign-off
Parallel Track A (RHEL + Podman stabilize)	App remediation, license purchase, DR execution

Master mapping: ENGAGEMENT-ALIGNMENT.md — SOW contract baseline. Original contract: SOW.md.

When client network is not in IaC

Typical for this engagement: Hub, spoke, firewall, peering, NSG, and private endpoints are built and changed via **Azure Portal, Firewall Manager, tickets, or vendor runbooks**—not Terraform/Bicep in the client's (or this) repo. Workshop 1 also noted **manual** server provisioning.

That is **expected**. It does **not** block discovery or the RHEL uplift strategy (STRATEGY-POST-RHEL.md).

What the consultant does instead

Activity	Use as input	Do not do
Discovery	Visio/diagram, IPAM spreadsheet, firewall export, portal screenshots, change tickets	Wait for network Terraform or recreate hub in this repo
Gaps	Risk register + recommendations with owner (Network, DNS, Platform)	Imply consultant will apply rules in production
Standards	"RHEL/Podman workloads require these network controls"	Mandate client adopt Terraform for network
Reference IaC in repo	Build spoke, SIG, VM patterns for platform team	Replace client's entire landing zone

Classify every discovery finding

Bucket	Example	Consultant output	Implementer
Platform IaC	VM, disk, SIG, image-build VNet module	Reference in repo; client forks	Client platform
Network config (manual)	Missing UDR, firewall deny, wrong PE DNS link	Allow-list doc, annotated diagram, ticket text	Client network
Process	6-week CAB, no IPAM owner	Risk + RACI	Client PM / network lead
Architecture	Flat VNet, spoke-to-spoke	Target-state memo (AZURE-NETWORK-ARCHITECTURE.md)	Client architect
App / data path	Snowflake, trading platform egress	Dependency in cutover plan	App + network
Out of SOW	Greenfield hub, SOC, pen test	Deferred / change order	Client or vendor

Extra questions when network is manual

#	Scope	Question	Why it matters
M1	Client-owned	How are VNets, peerings, NSGs, and firewall rules documented and changed today (portal, vendor, ticket system)?	Defines how recommendations are delivered
M2	Client-owned	Who can provide an as-built export (diagram, IPAM, rule matrix) without IaC?	Discovery blocker
M3	In scope	For each RHEL change (build, UAT, patch), what is the network change ticket template and SLA?	Schedules migration
M4	Advisory	Which network changes are blockers vs nice-to-have for UAT/prod?	Prioritized backlog
M5	Out of scope	Will client adopt network IaC later?	Optional future-state note only —not consultant build

Finding log template (copy per item)

ID:
 Title:
 Bucket: Platform IaC | Network manual | Process | Architecture | App | Out of SOW
 Blocks: UAT | Prod | Image build | None
 Recommendation:
 Owner (team):
 Client ticket #:
 Consultant doc reference:

Client-facing line

"We are not rebuilding your network in Terraform. We need your as-built and network changes as client-owned work, while we standardize RHEL and Podman on top of the network you already have."

1. Engagement alignment

#	Scope	Question	Why it matters
1.1	In scope	Who is the primary decision-maker for network changes (CAB approver)?	SOW requires client coordinator.
1.2	In scope	Agree written RACI : network vs platform vs app vs consultant for migration?	Prevents scope creep (SOW: no app remediation).
1.3	In scope	Which environments are in scope (prod, DR, UAT, dev)? Freeze / blackout dates?	Migration planning (SOW item 1).
1.4	Client-owned	Provide existing documentation (IPAM, firewall matrix, diagrams, IaC repos).	Client access obligation; consultant consumes.
1.5	In scope	What network-level success criteria define "migration complete" for each wave?	Aligns cutover support (SOW item 5).
1.6	Out of scope	Business application acceptance criteria (functional test cases)?	App team — not network discovery.

2. Azure landing zone & topology

#	Scope	Question	Why it matters
2.1	In scope	As-built model: hub-spoke, vWAN, or flat VNets? Where will migrated RHEL and image-build attach?	Planning + standards (SOW items 1–2).
2.2	Advisory	Address space standards and a non-overlapping block for image-build spoke (e.g. /26).	Recommendation; client allocates in IPAM.
2.3	In scope	Spoke-to-spoke allowed or hub-only east-west?	Risk + standards documentation.
2.4	Client-owned	Approved regions and data residency (US-only, etc.)?	Client policy; consultant designs within constraints.
2.5	In scope	Existing Azure Policy affecting VMs (no public IP, Trusted Launch, Image Builder)?	Risk identification (SOW item 3).
2.6	Out of scope	Greenfield design and build of entire hub/spoke landing zone?	Only recommendations unless change order; client implements .
2.7	Out of scope	Procurement of ExpressRoute circuits or hardware?	Client / telecom vendor.

3. Hybrid connectivity & egress

#	Scope	Question	Why it matters
3.1	Client-owned	ExpressRoute / VPN as-built; who owns connectivity subscription?	Consultant needs facts; client operates .
3.2	In scope	Gateway transit on hub peering for spokes?	Blocks hybrid access if wrong.
3.3	In scope	Default egress path (UDR → firewall/NVA vs direct)?	Image build + patching risk (SOW item 3).
3.4	Advisory	Draft firewall allow-list for RHEL build/patch (Red Hat, Microsoft, storage, AIB).	Consultant recommends; client implements rules.
3.5	In scope	Mandatory HTTP proxy? PAC URL and required exceptions?	Migration / image-build risk.
3.6	In scope	Forced tunneling to on-prem for all Internet?	Risk if build/patch paths unclear.
3.7	Out of scope	Implementing or tuning Azure Firewall / NVA rulebases long-term?	Client network ops (consultant advises only).

4. Golden image build network (Azure Image Builder)

Reference IaC: *ARCHITECTURE.md* — illustrative; deployment is **client-owned** unless contracted.

#	Scope	Question	Why it matters
4.1	Advisory	Dedicated build spoke vs shared CI VNet?	Standard recommendation (SOW item 2).
4.2	Client-owned	Approve subnet for AIB staging (e.g. /26) and peering/UDR implementation?	Client provisions network.
4.3	In scope	Inbound policy for build subnet (deny-all recommended)?	Hardening standard.
4.4	Advisory	Private endpoint for build script storage + DNS linkage approved?	Environment recommendation (SOW item 4).
4.5	Client-owned	Who owns Private DNS zones (<code>privatelink.blob</code> , <code>privatelink.vaultcore</code>) and VNet links?	Client DNS team executes.
4.6	Client-owned	Is Microsoft.VirtualMachineImages registered and allowed by policy?	Client subscription governance.
4.7	Client-owned	CAB process for firewall/peering/PE changes; lead time?	Client change management.
4.8	Out of scope	Consultant terraform apply in client production subscription?	Reference code in repo only unless change order.
4.9	Out of scope	Operating image pipeline on-call 24x7 after KT?	Client platform ops.

5. DNS, NTP, and identity

#	Scope	Question	Why it matters
5.1	In scope	DNS model for RHEL VMs (Azure, resolver, on-prem AD forward)?	Golden image + migration planning.
5.2	In scope	Required NTP sources for CIS/compliance?	RHEL standard (SOW item 2).
5.3	Client-owned	AD / Entra join required (SSSD)? OU, GPO, Kerberos constraints?	Identity team ; consultant captures requirements for image.
5.4	Advisory	PTR / reverse DNS requirements?	Recommend in standards; client DNS implements.
5.5	Out of scope	Designing enterprise Active Directory or Entra tenant?	Identity architecture — separate engagement.
5.6	Advisory	Developer CLI access — Entra group + PIM eligible roles on dev sub (Reader , AcrPush , KV Secrets User)? Max activation hours?	Developers use <code>az login</code> after PIM — DEVELOPER-AZURE-CLI-ACCESS.md; module <code>developer-cli-access</code> .
5.6b	Advisory	Consultant / network engineer portal read — PIM eligible Reader only?	Track B discovery — separate from developer CLI roles.
5.7	Client-owned	Access review / offboarding date when vendor engagement ends?	Remove PIM eligibility and guest accounts; audit trail for examiners.
5.8	Advisory	Entra ID SSH or Bastion for Linux jump — per-user only?	Aligns human Azure access with VM admin model; no shared <code>deploy</code> passwords.

6. Security, segmentation, and compliance

#	Scope	Question	Why it matters
6.1	In scope	Required OS baseline (CIS L1/L2 , STIG , internal standard)?	RHEL standards doc (SOW item 2).
6.2	Client-owned	FIPS 140 required for which workloads?	Client + app teams test compatibility (SOW assumption).
6.3	Advisory	NSG / ASG patterns per tier for RHEL spokes?	Standards recommendation; client implements.
6.4	In scope	Public IPs on VMs allowed? Bastion / jump model?	Migration admin access planning.
6.5	Out of scope	Deploying DDoS Standard on all VNets?	Client implements if policy requires.
6.6	Out of scope	Implementing micro-segmentation products (Illumio, NSX, etc.)?	Client security vendor scope.
6.7	Advisory	Evidence expectations (flow logs, firewall log retention) for audits?	Recommend; GRC/audit owns sign-off (COMPLIANCE.md is informational only).
6.8	Out of scope	Formal regulatory examination or PCI ROC attestation?	Client GRC / QSA.

7. Secrets, RHSM, and PaaS private access

#	Scope	Question	Why it matters
7.1	Client-owned	RHEL licensing model (BYOS, Satellite)? Who stores activation keys in Key Vault?	SOW: client obtains licensing.
7.2	Advisory	Network path for build VMs to Key Vault (PE, firewall)?	Recommendation; client implements.
7.3	Client-owned	Satellite / internal repo URLs and ports vs public CDN?	Client owns repo architecture.
7.4	Advisory	Managed identity pattern for RHEL VMs (per-app UAMI)?	Standards (SOW item 2); client implements RBAC.
7.5	Out of scope	Purchasing Red Hat subscriptions or legal review of license terms?	Client procurement / legal.

8. Migration & cutover (network impact)

#	Scope	Question	Why it matters
8.1	In scope	Same IP required vs new NIC / LB backend?	Cutover planning (SOW items 1 and 5).
8.2	Client-owned	Load balancer / App Gateway config changes—who executes?	Network/app teams; consultant supports (SOW item 5).
8.3	In scope	Firewall rule deltas when moving Ubuntu → RHEL (agents, ports)?	Risk identification (SOW item 3).
8.4	Advisory	Blue/green temporary subnets or parallel rules needed?	Recommendation; client implements.
8.5	In scope	Rollback network steps (DNS, LB)? RTO/RPO expectations?	Cutover support planning.
8.6	Client-owned	Approved maintenance window length per tier?	Client CAB / business.
8.7	Out of scope	Application load testing, performance tuning, or code fixes during cutover?	Application remediation — out of SOW.
8.8	Out of scope	Database migration or schema changes?	Out of OS migration scope.

9. Observability & operations

#	Scope	Question	Why it matters
9.1	Advisory	SIEM agents: install in golden image vs post-deploy? Required egress to collectors?	Standard recommendation; client SOC operates SIEM.
9.2	Out of scope	Deploying Sentinel/Splunk/QRadar or writing detection rules?	Client SOC / separate SOW.
9.3	Client-owned	Firewall ticket process and SLA for rule changes?	Client operations.
9.4	In scope	Escalation contacts for migration weekend (network L2/L3)?	Post-implementation support (SOW item 5).
9.5	Out of scope	24x7 NOC for client production after engagement?	Client ops unless MSSP contract.

10. Governance, change control, and knowledge transfer

#	Scope	Question	Why it matters
10.1	Client-owned	CAB cadence and lead time?	Client governance.
10.2	In scope	Who maintains network IaC after KT?	Support planning (SOW item 6).
10.3	In scope	Which runbooks to create/update (peering, PE, cutover, image rebuild)?	Knowledge transfer (SOW item 6).
10.4	Client-owned	Named successors for firewall policy and IPAM?	Client org design.
10.5	Out of scope	Staff augmentation (consultant embedded as network engineer for 12 months)?	Advisory SOW — not staff aug unless contracted.

11. Risk register inputs (in scope to document; client mitigates)

Consultant maintains **network-related risks** (SOW item 3). Client owns remediation.

Risk	Scope of consultant action
Firewall blocks image build / patch / RHSM	Document; advise allow-list
Private endpoint DNS not linked to build VNet	Document; advise fix
FIPS breaks applications	Flag only — app compatibility is client-owned
Forced tunneling blocks AIB endpoints	Document; advise path
Policy blocks Image Builder / Trusted Launch	Document; advise exception process
Overlapping CIDR / insufficient IP space	Document; advise IPAM change
CAB lead time exceeds migration timeline	Document; advise schedule
Expectation that consultant fixes application issues	Redirect — out of SOW
Network exists only in portal/tribal knowledge (no IaC)	Document as-built; do not block on Terraform
Discovery finds issues IaC cannot fix (process, architecture)	Classify per buckets above; recommendations not repo modules

12. Deliverables checklist (consultant vs client)

Deliverable	SOW item	Consultant	Client network / platform
Discovery summary & RACI	1	✓ Author	✓ Review / approve
RHEL-on-Azure network standards (draft)	2	✓ Author	✓ Approve / adopt
Network risk register	3	✓ Maintain	✓ Mitigate owned items
Environment recommendations (PE, build spoke, egress)	4	✓ Advise	✓ Implement
Cutover network runbook (draft)	5	✓ Author; support in window	✓ Execute
KT sessions & runbook handoff	6	✓ Deliver	✓ Attend; operate
Hub-spoke terraform apply in prod	—	× Unless CO	✓
Firewall rule implementation	—	× Advise only	✓
Application remediation	—	× Out of scope	✓ App teams
RHEL license purchase	—	× Out of scope	✓ Client

13. Questions to decline or redirect (out of scope)

Use this list when workshops drift. Polite redirect: *“That’s out of this engagement per SOW; we can note it as a dependency for [team].”*

- Fix application bugs, middleware config, or JVM/crypto stacks after FIPS
- Run QA/UAT test suites or sign off business functionality
- Buy RHEL subscriptions or negotiate Red Hat contracts
- Perform backups/restores or DR tests (client executes)
- Build greenfield hub, ExpressRoute, or firewall from scratch (without change order)
- Operate SIEM/SOC, write detections, or respond to incidents 24x7
- Provide legal, PCI, OCC, or audit attestation
- Penetration test or red team
- On-call as primary network engineer after KT period ends

Related project docs

Doc	Scope note
SOW.md	Original contract (items 1–6); charter aligns discovery
ARCHITECTURE.md	Reference golden-image design — not an implied deploy obligation
AZURE-NETWORK-ARCHITECTURE.md	Reference hub–spoke patterns
COMPLIANCE.md	Informational — not legal or audit sign-off
ENGAGEMENT-ALIGNMENT.md	Master scope — start here
STRATEGY-POST-RHEL.md	Phased plan; network = recommend, client implements